

Robust ArUco Marker Detection with Gamma Preprocessing and Learned False-Positive Filtering

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Abstract. This paper describes a practical pipeline for detecting ArUco fiducial markers in the FlyingArUco challenge setting, where markers are composited into cluttered natural images and must be reported by marker ID and top-left corner location. A direct OpenCV ArUco baseline gives a public test score of 0.783, but it produces many false positives. Since the competition metric penalizes false detections in the denominator, improving precision is as important as recovering missed markers. The proposed system keeps the detector itself simple: an inverse-gamma preprocessing step improves the input contrast for threshold-based detection, and a Gradient Boosting classifier removes false positives using 26 hand-crafted features extracted from geometry, intensity, binary cell structure, and image context. On the training set, the full pipeline improves the mean score from 0.780 to 0.952; on the public test set, it reaches 0.940. Ablation experiments show that multi-pass detection, relaxed error correction, and CNN-based recovery add more false detections than useful markers, making conservative post-filtering the more reliable strategy for this task.

Keywords: ArUco markers · fiducial detection · gamma correction · false-positive filtering · computer vision

1 Introduction

Fiducial markers are widely used in robotics, augmented reality, camera calibration, and pose tracking because they provide a compact visual code that can be detected and decoded from a single image. ArUco markers are especially useful in academic and industrial prototypes because OpenCV provides an efficient detector and several predefined marker dictionaries.

The challenge considered in this work is more difficult than detection in a controlled laboratory scene. The FlyingArUco dataset places markers on MS COCO backgrounds with changes in scale, perspective, texture, and illumination. The required output for each image is a list of marker IDs and the corresponding top-left corner coordinates. This means that a prediction can fail in two ways: by decoding the wrong ID, or by localizing the correct marker too far from the target corner.

The scoring function also changes the optimization target. For each image, correct predictions are rewarded by a Gaussian function of normalized localization error, while false positives increase the denominator:

$$\text{Score}_{\text{img}} = \frac{\sum \phi(d_{\text{norm}})}{N_{\text{gt}} + \lambda N_{\text{spam}}}, \quad \phi(d) = \exp\left(-\frac{d^2}{2\sigma^2}\right), \quad (1)$$

where $\sigma = 0.02$ and $\lambda = 1$. A method that recovers a small number of missed markers but introduces many false positives can therefore reduce the final score. This observation guided the design of the final pipeline.

The main contributions of this project are as follows. First, the effect of common preprocessing operations is measured systematically instead of selected heuristically. Second, the final detector combines OpenCV ArUco detection with a lightweight learned post-filter, avoiding unnecessary detector relaxation. Third, negative results from multi-pass detection and CNN recovery are analyzed to justify why the final method is conservative.

2 Related Work

ArUco detection. The standard ArUco pipeline detects candidate quadrilaterals, warps each candidate to a canonical square, thresholds the binary cells, and matches the decoded pattern against a dictionary [1]. OpenCV exposes this pipeline through `cv2.aruco.ArucoDetector`, including parameters for adaptive thresholding, perimeter constraints, error correction, and corner refinement.

Image enhancement. Classical enhancement methods such as CLAHE [3], histogram equalization, sharpening, and gamma correction are often used before threshold-based detectors. In this project, these methods are treated as measurable components rather than assumed improvements. Several common enhancements increased the number of false marker-like contours and reduced the score.

Learning-based fiducial detection. Recent approaches such as DeepArUco++ use neural networks to improve marker detection under difficult lighting and motion conditions [2]. The present work uses learning in a narrower role: a small classifier filters detections produced by a classical detector. This keeps the system easy to train and inspect while still addressing the main error mode.

3 Method

The final pipeline has three stages: preprocessing, ArUco detection, and post-filtering, as shown in Fig. 1. The detector uses the `DICT_ARUCO_MIP_36h12` dictionary and sub-pixel corner refinement. All other OpenCV detector parameters are left at their default values.

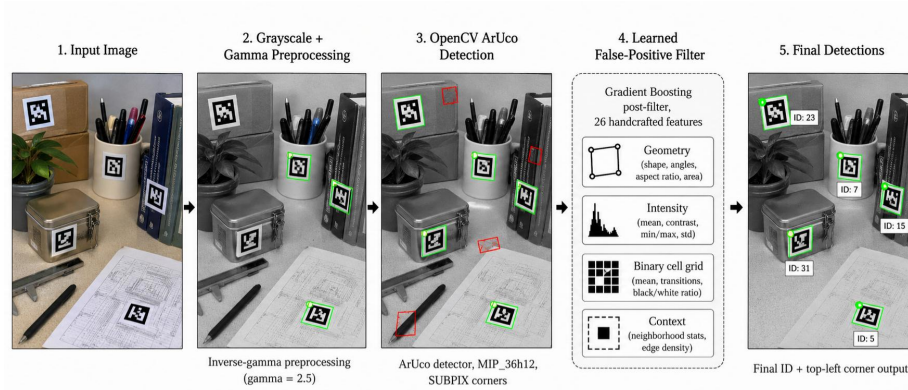


Fig. 1. Overview of the proposed detection pipeline. The image is first converted to grayscale and enhanced with inverse-gamma preprocessing. OpenCV ArUco detection then produces both valid marker candidates and false positives, which are removed by a learned post-filter before producing the final marker ID and top-left corner predictions.

Table 1. Main ablation results. Train scores are computed on the full training split; test scores are from the public Kaggle evaluation.

Pipeline	Train	Test	Spam	Missed	Gain
OpenCV baseline	0.780	0.783	1700	966	–
+ Gamma preprocessing	0.853	–	1322	508	+0.073
+ Geometric heuristic filter	0.890	–	697	508	+0.037
+ Learned post-filter	0.952	0.940	0	~480	+0.062

3.1 Dataset and Baseline

The training split contains 2000 images and 10,407 ground-truth markers, with an average of 5.2 markers per image. The baseline OpenCV detector produces 11,149 predictions on the training set. This is more than the number of ground-truth markers, so the baseline is not mainly limited by candidate generation. Its more important weakness is that it produces many incorrect marker IDs or spurious locations.

Table 1 summarizes the main pipeline stages. The baseline obtains 0.780 on the training set and 0.783 on the public test set. Adding preprocessing and filtering increases the training score to 0.952 and the public test score to 0.940.

3.2 Gamma Preprocessing

Each image is converted to grayscale and transformed using the following inverse-gamma lookup table:

$$I' = 255 \left(\frac{I}{255} \right)^{1/\gamma}. \quad (2)$$

Table 2. Selected preprocessing results on the training split. Each method is applied before the same OpenCV ArUco detector.

Method	Score	Gain	Spam	Missed IDs
Gamma 1.5	0.836	+0.056	1500	475
None	0.780	–	1700	966
Contrast stretch	0.772	-0.008	1826	955
Sharpen mild	0.743	-0.037	2069	1186
CLAHE	0.741	-0.039	2177	1123
Histogram equalization	0.730	-0.050	1763	1560
Bilateral filter	0.722	-0.058	1243	1913
Gamma 0.6	0.602	-0.178	1567	3172

Table 3. Gamma tuning with the geometric heuristic filter enabled.

γ	Score	Spam	Missed IDs	Predictions
1.0	0.822	949	997	10352
1.5	0.870	906	508	10808
2.0	0.886	806	441	10780
2.2	0.887	774	472	10714
2.5	0.890	697	508	10597
3.0	0.889	595	624	10377
3.5	0.877	515	808	10105
4.0	0.861	448	1053	9780

With this convention, $\gamma > 1$ increases mid-tone intensities while keeping black and white endpoints fixed. The best value found in the experiments is $\gamma = 2.5$.

This result is dataset-specific. Markers are placed on natural backgrounds, where medium-bright textured regions frequently create false quadrilateral candidates. Gamma correction changes the grayscale distribution before adaptive thresholding and improves the separability of marker borders and internal cells. Table 2 shows that gamma correction is the only preprocessing family that consistently improves the baseline. CLAHE, sharpening, histogram equalization, and strong morphological transforms increase false positives or remove useful marker contrast.

The gamma value was then tuned more finely with the geometric filter enabled. The best region is between 2.0 and 3.0, with a peak at 2.5. Larger values remove too much useful local contrast, reducing recall.

3.3 False-Positive Filtering

The first post-filter is geometric. A detection is removed when its perimeter is below 5% of the image diagonal or when the ratio between its shortest and longest sides is below 0.35. These thresholds reject small noisy candidates and highly elongated quadrilaterals.

Table 4. Top feature importances from the learned post-filter.

Rank	Feature	Importance
1	Cell uniformity	0.296
2	Inner cell standard deviation	0.131
3	Warped patch standard deviation	0.093
4	Warped patch minimum intensity	0.073
5	Border mean intensity	0.058
6	Area	0.037
7	Normalized side standard deviation	0.037
8	Warped patch mean intensity	0.032
9	Warped patch maximum intensity	0.032
10	Normalized area	0.032

The final post-filter is a Gradient Boosting classifier trained from detections on the training split. A detection is labeled positive if it has the correct marker ID and its top-left corner is within 5% of the image diagonal from a matching ground-truth marker. Otherwise, it is labeled negative. This produces 9,925 positive samples and 1,191 negative samples.

Each detection is represented by 26 features:

- geometric features: perimeter, area, side ratio, side length variance, angle statistics, and convexity;
- intensity features: mean, standard deviation, minimum, maximum, dark-pixel ratio, and contrast of the warped marker patch;
- binary-pattern features: statistics over an 8×8 canonical cell grid, including border darkness and cell uniformity;
- context features: number of detections in the image, nearest-neighbor distance, and relative area.

The most important features come from the canonical binary grid. This is expected: true ArUco markers contain a black border and clear binary cells, while many false positives are textured natural-image regions that only resemble quadrilaterals.

4 Experiments and Negative Results

Several alternatives were tested before selecting the final pipeline. The most consistent failure mode was false-positive growth. Table 5 summarizes the main negative results.

Relaxing `errorCorrectionRate` is especially risky. The MIP 36h12 dictionary has a minimum Hamming distance of 12. Increasing the correction rate toward 1.0 allows too many bit errors to be corrected, so many background quadrilaterals decode as valid marker IDs. Multi-pass merging has a similar problem: even if each pass recovers a few correct markers, it also contributes new false positives.

Table 5. Negative results. Methods that add candidates are harmful under the spam-penalizing metric.

Approach	Score	Spam	Main issue
Baseline	0.780	1700	Many false positives, but still usable
Relaxed multi-pass	0.003	>10k	High error correction accepts many non-marker patterns
Union of five passes	0.450	13614	Each preprocessing pass contributes additional false positives
Complement merge	0.630	5885	New false marker IDs bypass the complement rule
CNN recovery	0.887	999	Recovered candidates are too noisy for the metric

A CNN was also trained to classify rejected quadrilateral candidates. Although the validation accuracy reached 86.5%, the recovered detections were not precise enough for the competition metric. This suggests that, for this dataset and scoring rule, candidate recovery needs very high precision before it becomes useful.

5 Error Analysis

The remaining errors mostly fall into three categories. First, some markers have very low contrast against dark or textured backgrounds. Gamma correction helps many medium-bright images but cannot recover markers whose border and background are nearly indistinguishable. Second, severe perspective distortion can make the canonical warp unreliable, causing the bit pattern to be decoded incorrectly or rejected. Third, partial occlusion breaks the quadrilateral contour assumption used by the OpenCV detector.

The learned filter does not solve these recall failures; it mainly protects the score by rejecting false positives. This trade-off is appropriate for the current metric, but it also means that future recall improvements must be introduced carefully.

6 Conclusion

The final system improves ArUco marker detection by focusing on the actual dominant error mode: false positives. A simple OpenCV detector is retained, gamma preprocessing improves the detector input, and a Gradient Boosting post-filter removes false detections using interpretable marker-structure features. The result is a training score of 0.952 and a public test score of 0.940.

The main lesson is that detector recall should not be optimized independently from the evaluation metric. In this challenge, multi-pass and relaxed-decoding methods appear attractive because they find more candidates, but they reduce

the score once false positives are counted. A conservative detector with a strong post-filter is more effective and easier to justify.

Future work should focus on high-precision recovery of currently missed markers, especially under severe perspective distortion and low contrast. Better rejected-candidate labeling and explicit bit-pattern verification may improve recall without reintroducing the false-positive problem.

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